

# **CRT Chapter 13**

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1.

Stacks and arrays are both used to store multiple values in an ordered way. A stack is often implemented using an array, and both structures can be created with a fixed size depending on the program.

2.

Compilers use stacks to manage function calls and returns. When a function is called, the stack stores important information like local variables and where the program should return after the function finishes.

3.

The first output would be: 8 13

The second output would be: 8 12

4.

A stack is similar to a pile of plates because items are added and removed from the top. The plate that was placed on the stack most recently is the first one taken off, which follows the last-in, first-out (LIFO) rule.

5.

The first output would be: 5 8

The second output would be: 12 5

6.

**FIFO means that the item that enters the structure first is also the first one to leave. LIFO means that the most recently added item is the first one to be removed.**

**7.-Passengers waiting in line to get on a bus or airplane.  
-Cars lining up at a fast-food drive-through.**

**8.**

- a) False — A stack only has a top; it does not have a front or rear.**
- b) True — A stack can be completely emptied by repeatedly popping elements.**
- c) False — The top refers to the most recently added item, not the first one added.**
- d) False — isEmpty returns a true/false value, not a number.**
- e) True — A queue can hold many data items.**
- f) False — In a queue, elements are removed from the front, not the rear.**
- g) False — enqueue adds an item to the rear of the queue.**
- h) True — The first node in a linked list is called the head.**
- i) True — A node represents an element in a data structure, including stacks implemented with linked lists.**
- j) True — The total number of elements can be found using a size or length function.**
- k) True — In a linked list, the tail node points to null.**